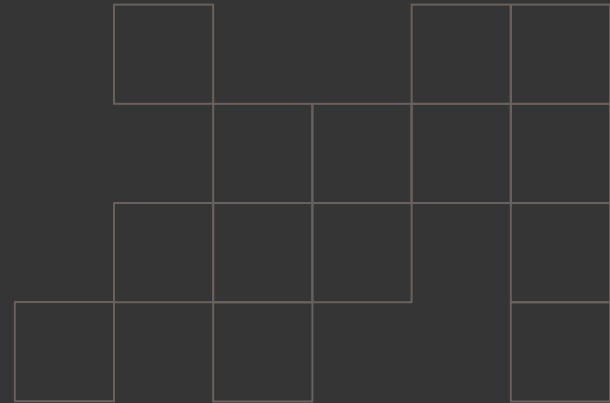
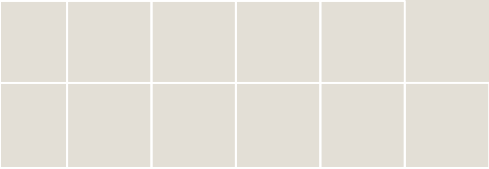


Team ZJL
10/5/2025

AI Tutor - Sprint 1

Demo and Presentation Video





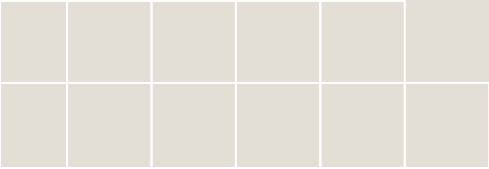
Introduction

Team Members:

Zachary Mullin, Jesse Boakye, Luis Zarate

Project: AI Tutoring System for schools





Sprint Objectives

The goals for Sprint 1:

- Create Core Documentation
- Collect Client Requirements
- Identify Full-Stack Framework
- Develop Initial Features





Requirements Gathering

The client gave us a lot of creative freedom with how we chose to design the project, but gave us two specific requirements:

- The AI tutor is an individualized system, with each student guiding their learning via their own personal tutor rather than a teacher guiding their learning.
- The system should run on a school district server, preferably with a local LLM to stay disconnected from the internet.

In addition, we discussed a couple other important requirements, including:

- The project must be secure and comply with federal law regarding privacy and security in education.
 - The AI needs to have guard-rails to prevent it from going off-topic or from being able to discuss inappropriate topics with students.
- 

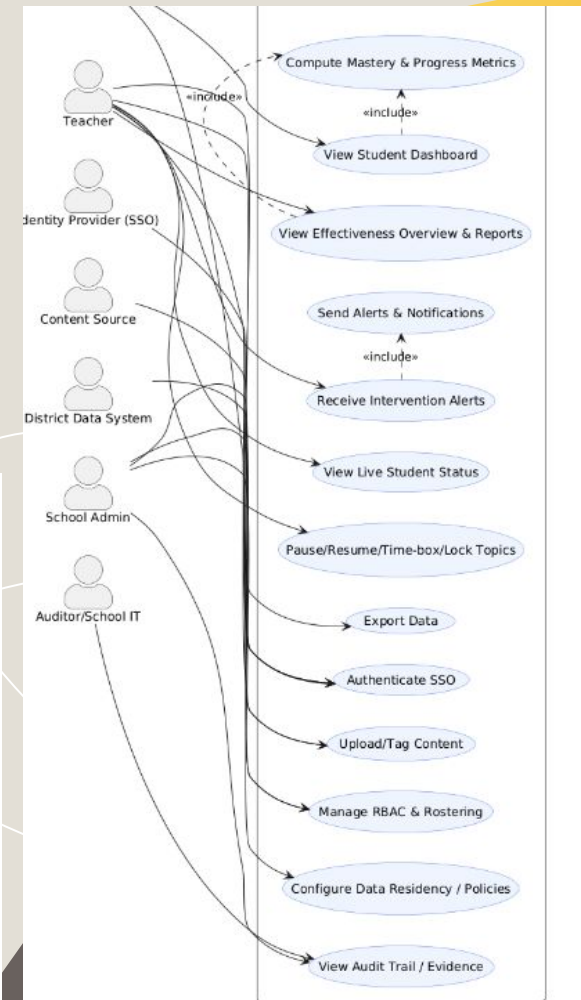
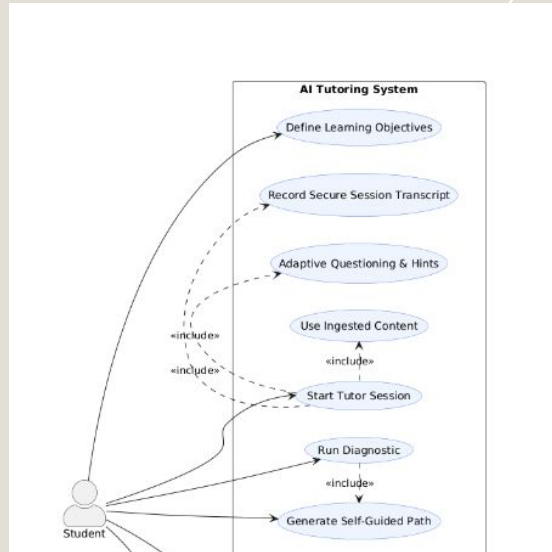


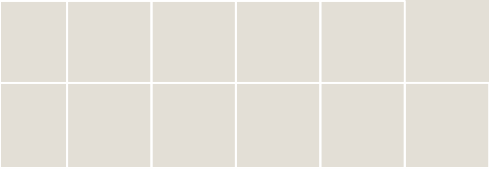
Documentation Of User Stories, Use Cases, and UML Diagrams

- I, as a teacher, should be alerted when a student is struggling even with the help of their AI tutor.
- I, a school district admin, should be able to deploy a local LLM and the AI tutor program on my school district server.
- As a school administrator, I want to review AI tutor usage reports so that I can evaluate adoption and effectiveness across different classrooms.

Documentation Of User Stories, Use Cases, and UML Diagrams

Use Case Diagrams

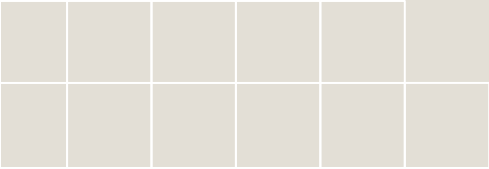




Installation and Evaluation of Existing Solution

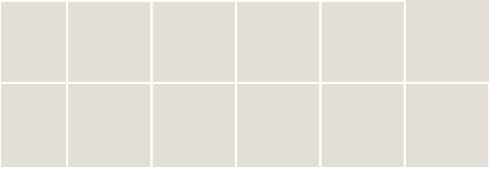
- Current methods of AI tutoring: MagicSchool, Khan Academy
- The problem: These are not individualistic
 - Teachers control the learning, and the students have to adhere to rigid curriculums
 - Students should get more say in how they use their tutors to learn





Kanban Overview

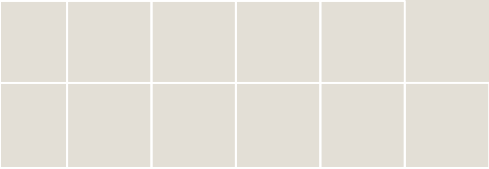




Feature Demonstration

- Implemented features:
 - Web based GUI for AI interaction.
 - Prompt safeguards ensuring safe responses
 - JSON and CSV based question and answer system for quizzes, to be fully implemented later.





Skills Identification and Learning Process

Zack:

- Django templates
- Gemini API
- Prompt Safety

Jesse:

- TailwindCSS design
- Environment configuration

Luis:

- JSON parsing
- Environment configuration



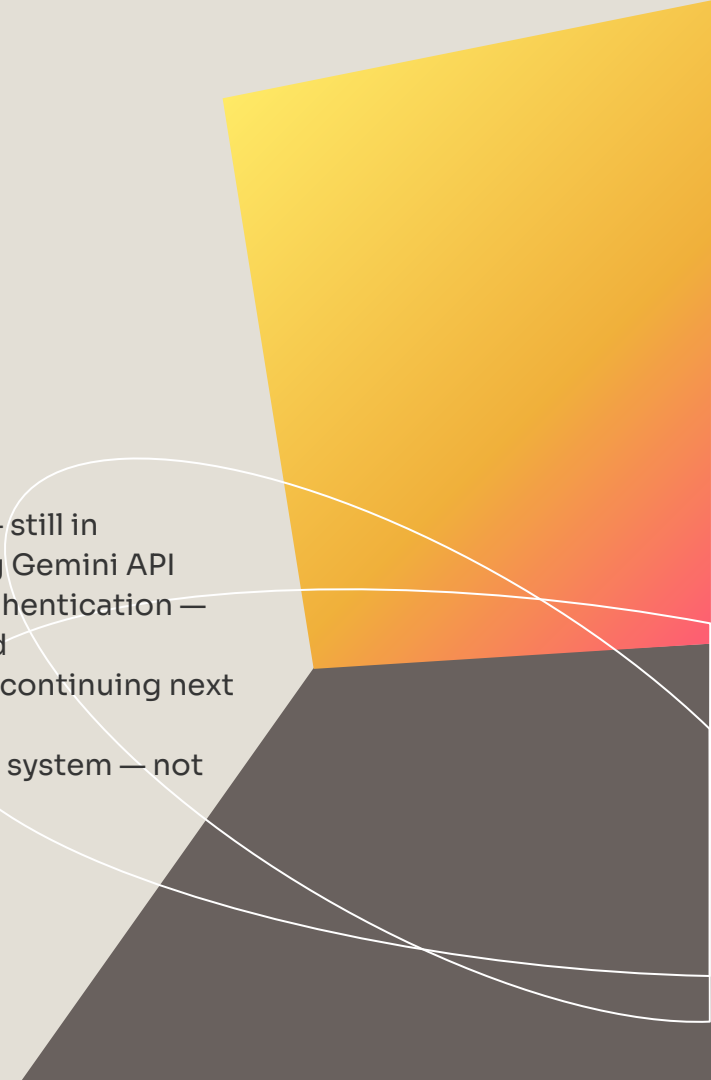


Sprint Achievements and Challenges

Sprint 1 Achievements

- Gathered client requirements & finalized documentation
- Created user stories, use cases, and UML diagrams
- Built early features:
 - Interactive AI tutor (Gemini API)
 - JSON quiz generation & grading
 - Web-based UI prototype
- Strengthened skills in prompt engineering

Challenges & Solutions

- Local LLM deployment — still in research; currently using Gemini API
 - Role-based access & authentication — database schema planned
 - Accessibility features — continuing next sprint
 - Backend logging & audit system — not yet integrated
- 

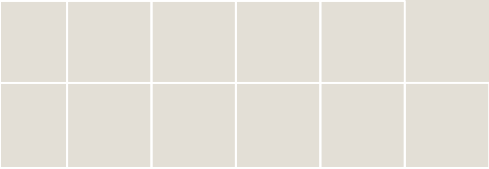


Responsible Use of AI

- Used ChatGPT to specify our project requirements
 - Improved our functional and non-functional requirements
- Used to Github Copilot and ChatGPT to assist in debugging the prototype

All final work and decisions are done by our team.

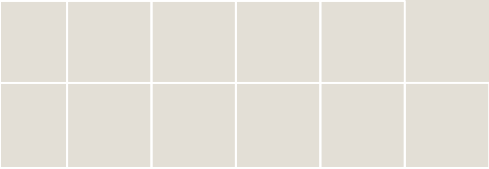




Next Steps and Sprint Retrospective

- Add teacher and admin dashboards
- Implement student progress tracking, and intervention alerts
- Develop role-based authentication and access control
- Implement the landing page, and login/sign-up page

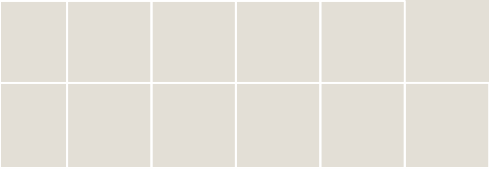




Conclusion

Thank you!





Youtube Video Link

https://www.youtube.com/watch?v=k_eLMM4389k

